Amir Beck, Chapter 1 - Exercise 1.1

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12 October 2026

1.1 Show that $||.||_{\frac{1}{2}}$ is not a norm.
By definition, for any $\mathbf{x} = (x_1, ..., x_n) \in \mathbb{R}^n$

$$||\mathbf{x}||_{1/2} = \left(\sum_{i=1}^{n} |x_i|^{1/2} \right)^2 .$$

Take $n = 2$ and $\mathbf{x} = (1, 1), \mathbf{y} = (1, 0) \in \mathbb{R}^2$. We have that

$$||\mathbf{x} + \mathbf{y}||_{1/2} = (\sqrt{|1 + 1|} + \sqrt{|1 + 0|})^2 = (\sqrt{2} + 1)^2 = 3 + 2\sqrt{2} > 5$$

and

$$||\mathbf{x}||_{1/2} + ||\mathbf{y}||_{1/2} = (\sqrt{|1|} + \sqrt{|1|})^2 + (\sqrt{|1|} + \sqrt{|0|})^2 = 4 + 1 = 5$$

That is,

$$||\mathbf{x} + \mathbf{y}||_{1/2} = 3 + 2\sqrt{2} > 5 = ||\mathbf{x}||_{1/2} + ||\mathbf{y}||_{1/2}$$

which means that $||.||_{1/2}$ does not satisfy the triangle innequality. Therefore $||.||_{1/2}$ is not a norm.